ON3VZ · A REPLY TO WIMO

Heart Pounding Before the First QSO

From a brand-new Class C callsign to pile-ups, split and Borneo: five months of discovering amateur radio, and what I would tell the next newcomer



ON3VZ at the operating position in Hoboken, Belgium.

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October 2026 · on3vz.github.io · qrz.com/db/ON3VZ

Why I am writing this

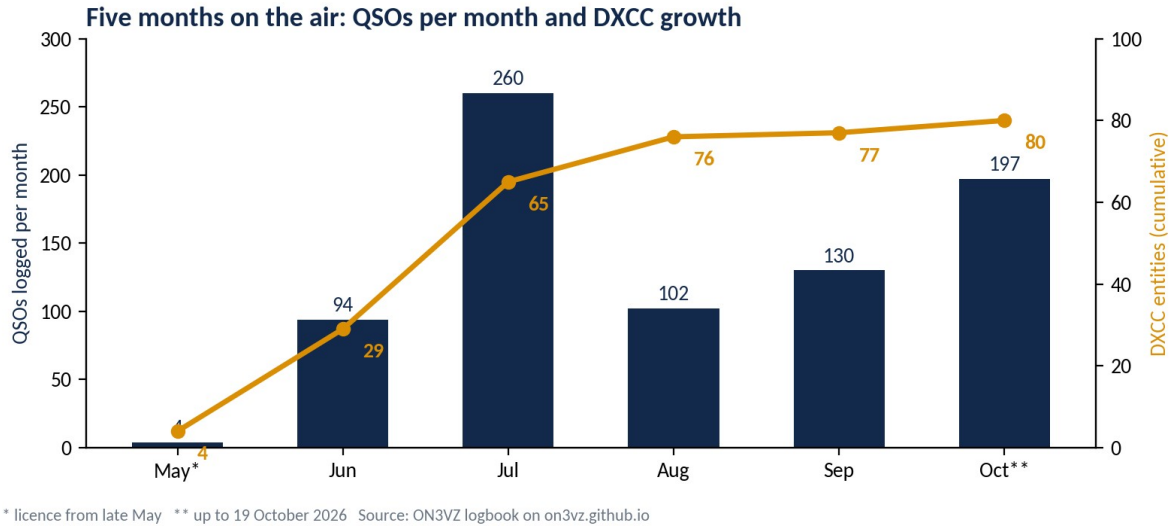
A few days ago WIMO asked a simple question in their newsletter: do you remember your first QSO? The hand on the microphone that trembles a little, the heart pounding in your throat, the cheat sheet with your own, still unfamiliar callsign within reach on the desk. They put together a five-step guide for newcomers and invited all of us to share our story: which equipment was in front of you, and who was your very first contact?

Yes, I remember. Very well, in fact, because it is not that long ago. My first QSO was on **27 May 2026**. Reading WIMO's question made me stop and look back at what has happened in the five months since: how much I have learned, how many things I got wrong first and then understood, and how this hobby simply keeps on giving. Every week there is something new to discover, to understand and to try.

So this is my answer to WIMO, and at the same time a wrap-up for myself. It is also written for everyone who is standing where I stood in May: a fresh licence in hand, a head full of exam theory, and one big question. **Now what?**

ON3VZ AT A GLANCE

Callsign	ON3VZ (portable: ON3VZ/P)
Licence	Belgian Class C: 25 W on HF, VHF and UHF up to 50 W
Location	Hoboken, Antwerp, Belgium, grid JO21EE
First QSO	27 May 2026, LZ100LZ (Bulgaria), 20 m SSB
Home station	Icom IC-7300 MkII, RF.Guru IronWave 6 vertical, HyEndFed end-fed sloper
Portable / mobile	QRP Labs QMX 5 W · Yaesu FTM-510DE · Yaesu FT-65 · Baofeng DM-1801 Pro
Log so far	Close to 800 QSOs, about 80 DXCC entities, all six continents (to 19 October 2026)
Club	WLD (ON6WL), section of the UBA



Logged QSOs per month and the cumulative number of DXCC entities.

1. The licence, the callsign, and then…

Passing the exam feels like finally getting the key to a door you have been staring at for months. Then the callsign arrives, **ON3VZ**, and reality sets in. I knew the theory. I did not know which radio to buy, which antenna would work in a city garden, which software to install, or what a real QSO actually feels like.

What followed was a real search. Evenings full of reviews, forum threads that contradict each other, shop pages, manuals and videos. I wanted the right equipment, and I wanted to do it properly: within the rules of my Class C licence, safe, clean, without bothering the neighbours, and with a station that performs as well as physics allows at 25 W.

One decision I made early on shaped everything that came after: **I would document everything on my own website and blog.** That documentation has two purposes, and both matter to me equally.

- **For myself.** Writing something down forces you to really understand it. My website is my notebook, my reference and my memory: which settings I used, which choices I made and, above all, why. When I want to know how I configured something three months ago, I look it up on my own site.
- **For others.** Every time I searched for an answer, I thought: the next newcomer should not have to search this long. So I write my posts for the person who is just starting out and looking for the same answers I was looking for. If one beginner finds a clear explanation there and makes a good first QSO a little sooner, it was worth the effort.

That is how on3vz.github.io grew into what it is today: an interactive logbook with a world map, a DXCC matrix, my shack, mobile and portable stations, a QSL wall, awards, and a blog where I explain my choices, the settings and the reasons behind them, step by step and with the "how" included.

> **THE CLUB CAME FIRST**
>
> My amateur radio story did not start with the licence. It started with the club. Before I was licensed, I joined a virtual fox hunt and an ARDF training day with radio club WLD (ON6WL). Somewhere between tracking fox transmitters and tuning antennas, the experienced members kept talking about CW with a quiet enthusiasm that was impossible to ignore. I drove home thinking: I want to learn that. More about that later.

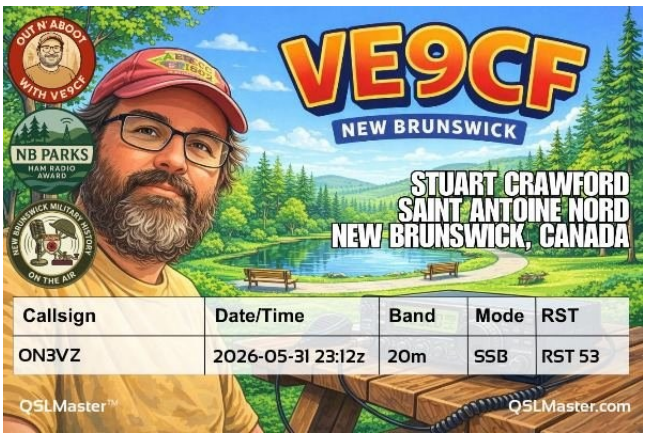
2. The first QSO

Wednesday 27 May 2026, 14:39 UTC, 20 metres, SSB. The station on the frequency was **LZ100LZ**, a special event station in Bulgaria. Not a quiet station waiting for a chat, but one with a pile-up in front of it.

I had rehearsed my phonetics, Oscar November Three Victor Zulu, about two hundred times in my head. My voice still cracked, and I still got it wrong on the first try. So I listened, waited for a gap, gave my callsign once, as cleanly as I could, and waited. And he came back to me. **57 sent, 59 received.** My first QSO, through a pile-up, logged.

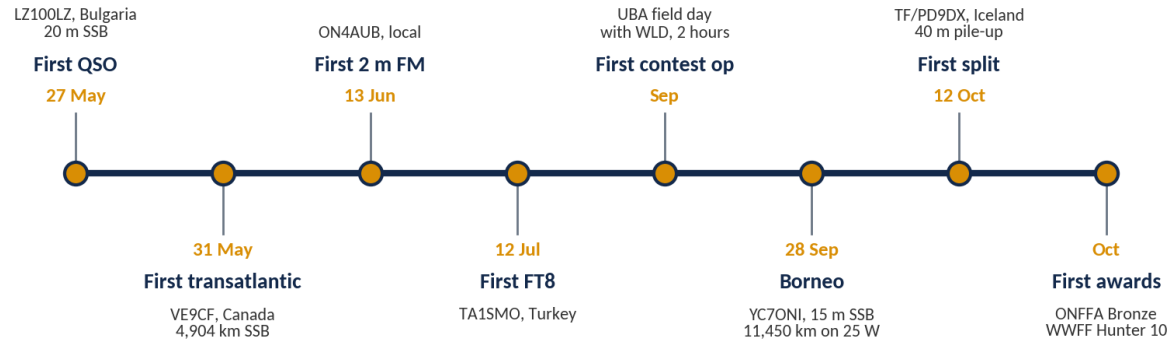
To answer WIMO's question directly: in front of me was an **Icom IC-7300 MkII** at 25 W, connected to an **RF.Guru IronWave 6** multiband vertical in the middle of the garden. The other station was LZ100LZ in Bulgaria. Two more special event stations followed that same day, in Russia and Croatia.

Four days later, at 23:13 UTC on 31 May, **VE9CF** in New Brunswick, Canada came back to me on 20 m SSB. 4,904 km, on 25 W, four days into my licence. His colourful eQSL card still has a place on my QSL wall.



The eQSL from VE9CF, Canada, 31 May 2026: my first contact across the Atlantic, four days after my first QSO.

A timeline of firsts, 2026



My firsts in 2026, from the first QSO to the first awards.

3. The search for the right equipment

The transceiver

I compared several radios, among them the Yaesu FT-710 Field, and chose the **Icom IC-7300 MkII** for the home station. For a Class C operator it fits very well: a built-in antenna tuner, a real-time spectrum scope that shows at a glance which bands are alive, a receiver that copes well with urban noise, and a power setting that sits naturally at the 25 W I am allowed to use.

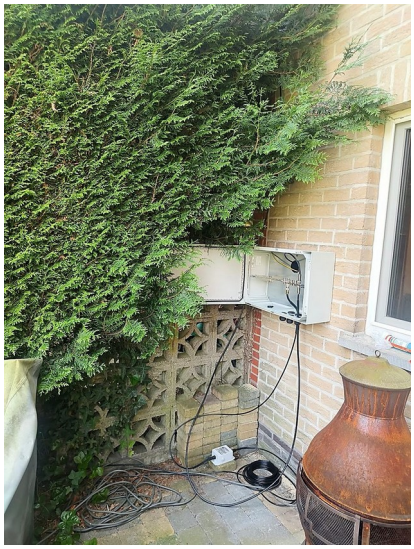
Why one antenna is never enough

The first hard lesson was this: **no single antenna does everything well**. That is not a marketing story, it is physics. A vertical radiates at low angles, which is what you want for long distance (DX). A horizontal or sloping wire at modest height sends more energy upwards, which is what you need for NVIS, the short-range skywave that covers a few hundred kilometres on 40 and 80 m. An antenna that promises to do both usually does neither particularly well.

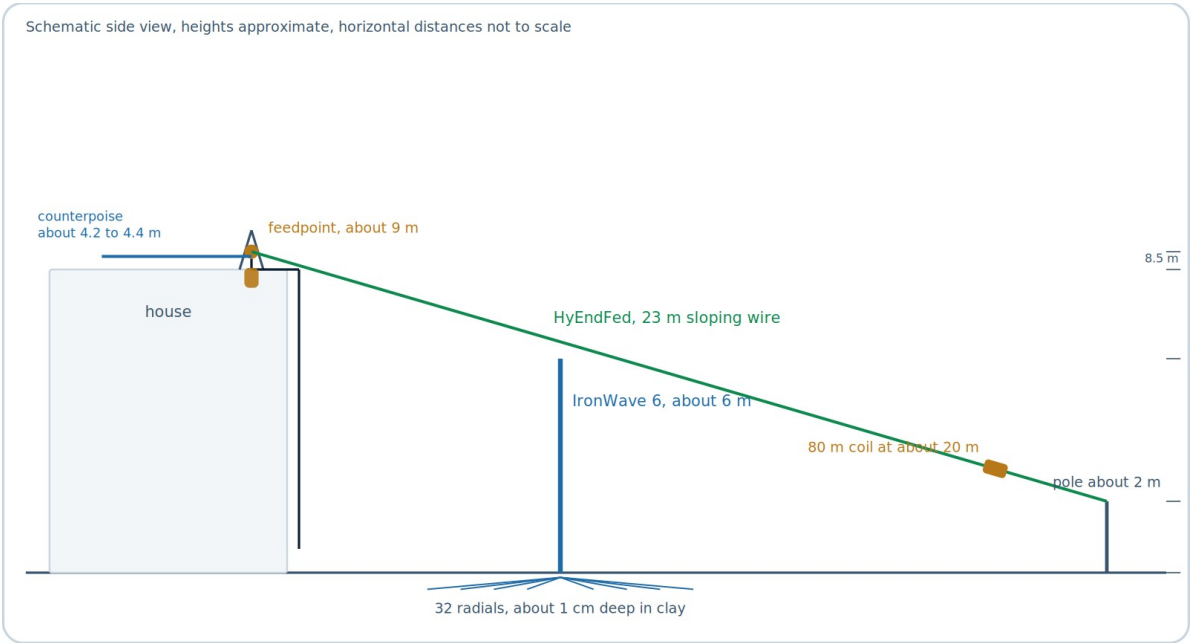
My situation added its own constraints: a city garden, neighbours close by on all sides, a solar inverter running most of the day, and a partner with a firm opinion about how many antenna structures were acceptable on the roof. The answer became two antennas, each with its own job:

- **RF.Guru IronWave 6**: a 6 m vertical with a 4:1 transformer at the base and **32 radials** buried in the lawn. My DX antenna for 20, 15 and 10 m, and 30 m with extra tuning.
- **HyEndFed 5-band end-fed wire** (23 m) as a sloper: the feedpoint about 9 m high on the flat roof, sloping down over the garden to a 2 m pole. My antenna for 40 and 80 m, with NVIS for the regional contacts. It is a temporary solution until a dedicated low-band antenna arrives, after which it goes back into the portable kit.
- Both antennas run to an **outdoor antenna cabinet** with an antenna switch, and into the shack from there.





The IronWave 6 standing free in the middle of a small city garden, and the outdoor antenna cabinet where the feedlines come together.



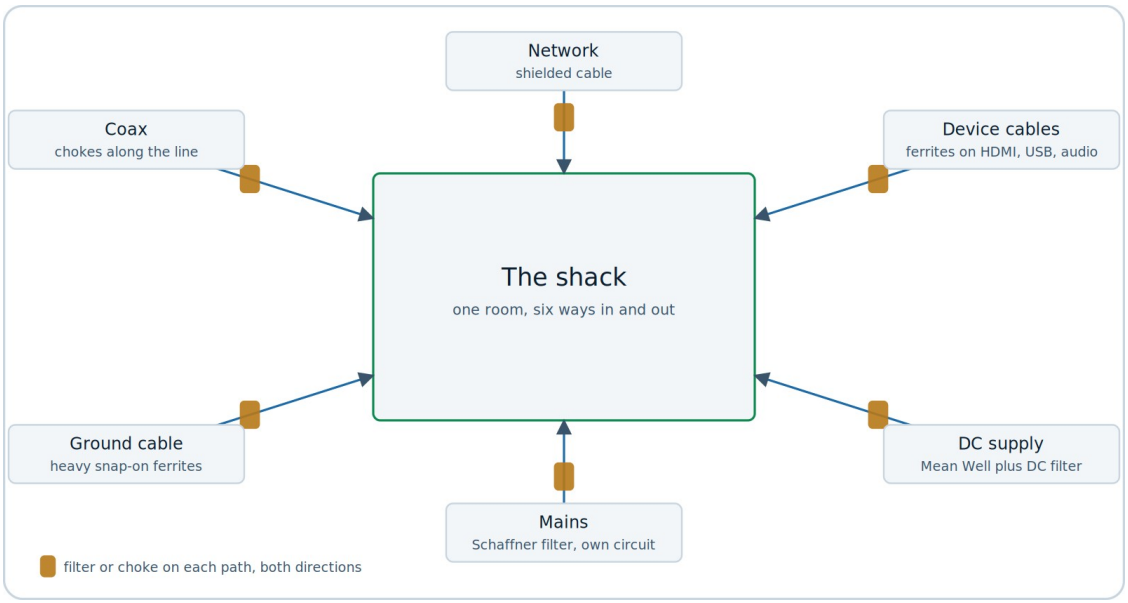
Side view of the two antennas: the HyEndFed sloping from the roof, the IronWave 6 with its radials in the garden.

I did not figure this out alone. **Joeri Van Dooren, ON6URE, of RF.Guru** guided me through the whole station: the antenna choice, the feedline, the chokes, the bonding and the installation. He even wrote up my first HF station as a case on the RF.Guru website: First HF station: antenna, bonding, chokes and SWR. If you are setting up your first station, it is well worth a read. The full story of my choices is in my post Starting from Zero: Building a Capable HF Station on 25 Watts.

4. Getting the shack right

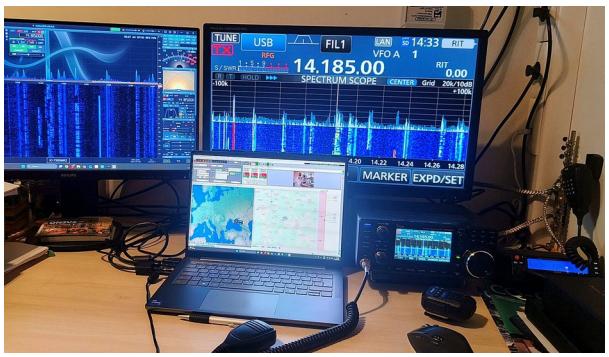
I wanted the shack to be right from day one, for DX as well as for NVIS, and without interference in either direction. The most important thing I learned is to think in **current paths**. Every cable that enters or leaves the shack is a path for RF and noise, in both directions. My shack has six of those "doors", and each one has its own choke or filter:

- **The coax**: common-mode chokes at the feedpoint, along the line, before the antenna cabinet and after the antenna switch.
- **The ground cable** from the antenna cabinet: wound through heavy snap-on ferrites.
- **The mains**: a separate shack circuit with its own Schaffner filter and star-wired outlets, plus a second Schaffner filter on the solar inverter.
- **The DC supply**: a low-noise Mean Well power supply followed by an RF.Guru DC noise filter.
- **The device cables**: ferrites on HDMI, USB and audio leads, everything bonded to one copper busbar.
- **The network**: a shielded network cable.



The shack as a room with six doors, each with its own filter or choke.

The result surprised me. Stations regularly tell me, without being asked, that I sound clean and loud. "It sounds like you are standing next to me." A station in Oman picked me out of a pile-up because I came through clearly over the other callers. On 25 W, from a city garden. There is no single trick behind it: it is the combination of many modest measures that makes the difference. I wrote all of it down, with the theory and my own calculations, in [When Theory Meets Practice: How a Layered Setup Keeps My 25 W Clean].





The operating desk, and one of the two Schaffner mains filters.

5. Setting up the radio and the computer

A modern station is half radio, half computer. These are the settings that made the biggest difference for me, with how to do them:

- **One USB cable, two ports.** The IC-7300 MkII shows up in Windows with two virtual COM ports. I give one to the logbook program for CAT control (frequency and mode follow the radio automatically) and the other to WSJT-X for FT8. Check the port numbers in Windows Device Manager under "Ports (COM & LPT)", and always plug the radio into the same USB socket on your PC so the numbers do not change.
- **Match the CI-V settings.** The baud rate and CI-V address in your software must match the radio. On the IC-7300 MkII you find them under MENU, SET, Connectors, CI-V. If the software cannot read the frequency, this is the first place to look.
- **Watch the Windows audio.** At one point I had 0 W output and no audio in FT8. The cause was Windows itself, blocking the radio's USB audio device. In Windows Sound settings, select the radio's "USB Audio CODEC" as the input and output device in WSJT-X, and make sure no other application has taken exclusive control of it.
- **Clean audio for SSB.** I set the TX equaliser with the bass kept low and the treble slightly up, use no speech compression, and keep the ALC calm. A quiet ALC meter is your friend: if it swings hard, you are overdriving and your signal gets wider and dirtier.
- **Time matters.** Let your PC synchronise its clock automatically. FT8 needs your clock within about a second, and every logbook entry is in UTC.

6. Choosing a logbook

There are many logging programs, and you can lose weeks comparing them. I compared Log4OM and DXLab Suite and chose **Log4OM v2**: free, widely used in Europe, and well integrated with all the confirmation services. My logging chain now looks like this:

- **Log4OM v2**: the central logbook at home, with uploads to **QRZ.com, LoTW, Club Log, eQSL and QRZCQ**.
- **WSJT-X** for FT8 and FT4, which sends every completed QSO straight to Log4OM over UDP.
- **N1MM Logger+** for contests and field days.
- **PoLo** on a tablet or phone for portable SSB, exported as ADIF and imported into Log4OM afterwards.

- **Two configurations** in Log4OM, one for ON3VZ and one for ON3VZ/P, so portable contacts land in the right place with the right callsign and location.

Three things I would tell any newcomer: log every QSO immediately and always in UTC, apply for your LoTW certificate early (in Belgium via the UBA), and make regular ADIF backups. Your log is your life's work as a radio amateur. It also pays off quickly: within eight weeks, 99 of my contacts were already confirmed. And because my website reads the same ADIF files, the logbook map, the DXCC matrix and the QSO dashboard update themselves. Everything I learned about logging and QSL cards is in From First QSO to First QSL Card.

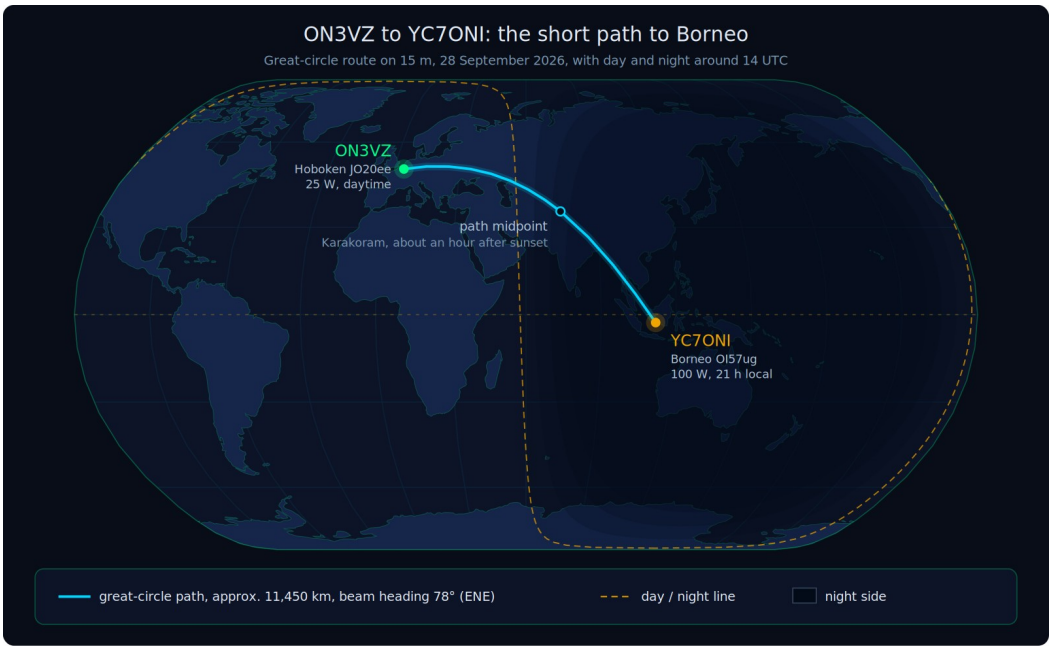
7. Learning to understand propagation

In the beginning, band choice was guesswork. I called CQ on whatever band was quiet and wondered why nobody answered. Quiet usually means closed. After eight weeks I analysed my first 247 contacts, and one number jumped out: **42 percent of them were made around twilight**, the grey line. During those hours the absorbing D layer has collapsed while the F layer is still charged, and a 25 W signal suddenly reaches much further. My practical model became: 20 m is home, 15 and 10 m are the daylight bonus, and 40 and 80 m were still homework. That analysis is in Eight Weeks On The Air: What 247 QSOs Taught a Class C Beginner.

Borneo on 25 watts

The moment propagation really clicked was on 28 September 2026. Less than a week after the autumn equinox, I worked out the conditions for a path to South-East Asia myself: the sunspot number, the predicted MUF (just over 22 MHz) and the path reliability, hour by hour. On paper the path to Borneo had only a **35 percent** chance of being open. Marginal, but not impossible.

So I kept listening. **Herman, YC7ONI**, on Borneo was fading in and out with deep QSB. Then his signal slowly built up. I followed his other contacts until I could copy them, sat up straight, called, and he came back to me. 15 m SSB, **53 sent, 53 received, about 11,450 km**, my 25 W against his 100 W. That is a difference of only 6 dB, about one S-unit. Planned, not lucky. The full story is in 25 Watts to Borneo.



The short path from Hoboken to YC7ONI on Borneo, about 11,450 km, with the night side shaded.

8. Measuring is knowing

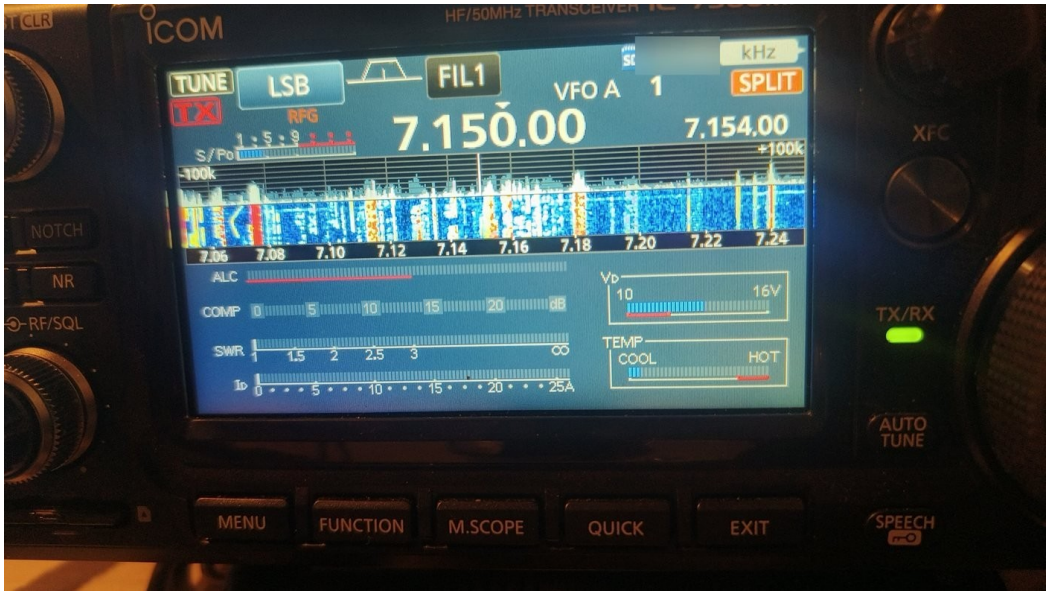
There is a Dutch saying that I have taken to heart: meten is weten, measuring is knowing. An antenna that seems fine can be resonant outside the band. A connector can have water in it. A small transmitter can put part of its power into a harmonic. None of that is visible, but all of it is measurable.

After a club workshop where half the room had a NanoVNA and nobody was quite sure what the numbers meant, I wrote a guide to decibels, and then bought a **NanoVNA-H4** and a **tinySA** myself. I measured the harmonics of my handheld on a spectrum analyser, swept both home antennas from 1 to 30 MHz, and finally saw on a screen why the IronWave is my DX antenna and the HyEndFed my low-band antenna. Measuring turned "I think" into "I know". My beginner's guide, with downloadable manuals in English and Dutch, is Measuring is Knowing: the NanoVNA-H4 and tinySA.

9. Understanding why: my first split

Some things you only really understand once you live them. On 12 October 2026 I heard it for the first time: a station calling "CQ CQ, listening on 7.154", and later "listening 4 up". **TF/PD9DX**, Derk, a Dutch operator in Iceland, with a huge pile-up in front of him on 40 m.

Split operation means transmitting on one frequency and listening on another, so the DX keeps his own frequency clean and can pick callsigns out of the crowd. The rule is simple: **you receive where the DX transmits, and you transmit where the DX listens.** On my IC-7300 MkII I put VFO A on 7.150 to hear him, set VFO B to 7.154 to call him, and pressed the SPLIT key until the orange SPLIT label lit up. I listened first, learned his rhythm, checked that my transmit frequency was clear, and called once. **59 both ways.** Iceland on 40 m from a city garden.



The split setting on my IC-7300 MkII: VFO A on 7.150 (receive), 7.154 on the right (transmit), and the orange SPLIT label on.

The lessons, written down while they were fresh: check that SPLIT is on before you call, call with your callsign only, never call while the DX is working someone else, switch SPLIT off afterwards so you do not disturb the next frequency, and log both the transmit and the receive frequency. All of it is in My First Split QSO.

Pile-ups on 25 watts

My very first QSO already went through a pile-up, and that set the tone. With 25 W you are not going to win a pile-up by force. What you can do is pick your moments: tune slowly, listen, learn how the DX works, wait for the gap, and give your callsign once, cleanly, with correct phonetics. Calling twice over someone else gets you nowhere. Combine that with a clean signal, and you will be surprised how often you get picked out.

10. Contests and field days

Contests turned out to be one of the best schools there is. You learn to listen for a callsign in the noise, to keep exchanges short, and to operate with discipline. My log now contains contacts from the IARU HF Championship, the RSGB IOTA contest, the UBA DX contest, the BMA 80 m contest and the UBA ON Contest.

Before my first field day with WLD I wanted to contribute something, and a newcomer can always contribute something. Club members kept mentioning the paper grid of which participating stations had already been worked on which band, so I built the **N1MM Field Day Tracker**: a live matrix fed straight from N1MM Logger+. Read about it in Building the N1MM Field Day Tracker.

Then came the **UBA Field Day** with WLD in September, and with it a real first: for the first time I sat **two hours as an operator in a contest**. With the club's log running and club members around me, my heart was pounding just as hard as on 27 May.



Paper proof of a busy autumn: place 35 in the UBA ON Contest 2026 (40/80 m SSB, ON stations), and my WWFF Hunter 10 certificate.

The certificates followed: place 35 in the **UBA ON Contest 2026** (40/80 m SSB, ON stations, 1,650 points), position 1,111 of 35,815 in the **WWA Sprint 2026** with 23 special event stations on three bands, an **ONFFA Bronze** award for 10 Belgian Flora & Fauna references, the **WWFF Hunter 10** certificate, and a commemorative **TOTA** certificate. They are all on my Awards & Certificates page.

11. Mobile, portable, DMR and digital

Mobile

In the car there is a **Yaesu FTM-510DE** for 2 m and 70 cm: 50 W, APRS and C4FM, powered by its own LiFePO4 battery so nothing is wired into the car's electrical system, with a Comet antenna on a magnet

mount. In my pocket there is a **Yaesu FT-65** handheld, which I use on the WLD club frequency. See my Mobile page.

Portable: ON3VZ/P

Once the shack was in order, I wanted to take a station outside, both for our holiday house in the Belgian Ardennes and for Flora & Fauna activations. I built **ON3VZ/P** around a **QRP Labs QMX** at 5 W and a **SOTAbeams Band Hopper IV** linked dipole on a 10 m mast. Two disciplines I always follow: the QMX always runs at 11.50 V through a DC-DC buck converter and is never connected directly to the LiFePO4 battery, and the dipole always has the same number of links closed on both legs.

The station is only half the work. I set up ON3VZ/P on QRZ.com, LoTW, Club Log and QRZCQ, created a portable configuration in Log4OM and WSJT-X, and wrote a field checklist in English and Dutch. The whole build is in Building ON3VZ/P: a QRP Portable Station for Flora & Fauna.





ON3VZ/P in the field: the QRP Labs QMX on 5 W, and the Band Hopper IV dipole on its mast.

DMR and digital voice

FT8 and FT4 opened up the bands for me: after my first FT8 QSO on 12 July, seventeen new DXCC entities followed in eight days. Digital voice was the next step. A club evening at WLD, where Patrick ON4PN took us from ones and zeros to D-STAR, Fusion, DMR and FreeDV, gave me the background (Digital Voice Modes). Then I programmed a **Baofeng DM-1801 Pro** from scratch: a DMR ID, a BrandMeister account, a 200-channel codeplug in 31 zones, roaming and a Pi-Star hotspot at home. The two-part guide starts with DM-1801 Pro, Part 1: Setup.



The DM-1801 Pro, programmed for FM and DMR in Belgium.

12. What is next

This hobby does not end. It just keeps opening new doors. Here is where I am heading.

The short-term dream: QRP, low power and antennas

What fascinates me most right now is the gap between theory and practice. On paper, every antenna and every transceiver has its numbers: gain, take-off angle, efficiency, losses, receiver performance, output power. In the field, the ground, the height, the feedline, the weather, the band conditions and the operator all have their say. I want to find out for myself how far apart those two worlds really are.

So my dream for the near future is to **experiment with QRP and low power, and with different antennas**, to find the best possible setup for **Flora & Fauna activations** (WWFF and ONFF) and other outdoor activations. Which antenna gets the most out of 5 or 10 watts in a forest clearing? How much does a resonant wire gain over a compromise antenna with a coil? What does a higher mast, a better choke or a different feedline really change? Which transceiver hears best when the signals are weak?

I am still looking for the right way to do this: a method to compare setups fairly, with measurements before I go out, a fixed test routine in the field, and the logbook, the reports received and the reverse beacon and spotting networks as evidence afterwards. The ON3VZ/P station with the QMX and the Band Hopper is my starting point. So far I have been a hunter, with ONFFA Bronze and WWFF Hunter 10 as proof. Now I want to be on the other side of the pile-up. And of course, every experiment will be documented on my site, results that disappoint included.

The near future: CW

I have been training my ears with the Koch method since before my licence, and the very first CW contacts are now in my log. CW is where a low-power station becomes truly competitive, so it fits perfectly with the QRP experiments. It is my next big learning goal.

Within two years: HAREC

The longer-term goal is clear: a **full HAREC licence within two years**. More bands, more power when I need it, and above all more to learn. Strangely enough, the better I understand QRP, the more I look forward to it.

13. My tips for the next newcomer

> **SURROUND YOURSELF WITH EXPERIENCED CLUB MEMBERS**
>
> This is the most important tip of all. I am a member of radio club WLD (ON6WL), and I received tons of help there: advice, patience, honest answers to beginner questions, and encouragement after a bad session. Club evenings teach you things no manual covers. Join a club before you buy anything, and ask your questions out loud. Nobody minds that you are new.

- **Listen more than you transmit.** Your first hundred QSOs are a listening exercise.
- **Do it properly from the start.** Chokes, bonding and filtered power are not luxuries. They give you a clean signal, a quiet receiver and happy neighbours.
- **Use two antennas if you can:** a low angle for DX, a high angle for NVIS.
- **Log everything, immediately, in UTC,** and set up LoTW, QRZ.com and Club Log from day one.
- **Learn propagation.** Put the grey line in your calendar. The two hours around sunset are worth more than a whole afternoon.
- **Measure, do not guess.** A NanoVNA and a tinySA cost less than a good SWR meter and teach you more.
- **Understand why.** When something works or does not work, find out the reason. That is how split, pile-ups and propagation stop being a mystery.
- **Write it down.** A website, a blog or a simple notebook. You document it for yourself, and at the same time you help everyone who is looking for the same answers.
- **25 W is a constraint, not a handicap.** It makes you a better operator, because you cannot brute-force anything.
- **Keep experimenting.** The hobby rewards curiosity.

Back to WIMO's question

Do I remember my first QSO? Every second of it. An IC-7300 MkII, an IronWave 6 in the garden, 25 watts, a cracking voice, and LZ100LZ in Bulgaria coming back to me through a pile-up. Five months later, close to 800 contacts and about 80 DXCC entities further, my heart still beats a little faster when a weak station comes back to me. I hope it never stops doing that. And the next chapter is already waiting: QRP, antennas and the field.

Thank you to everyone at **WLD (ON6WL)** for the help and the patience, to **Joeri, ON6URE, of RF.Guru** for the guidance on my station, to every operator who came back to a weak signal from a small station in Hoboken, and to **WIMO** for asking the question that made me look back.

See you on the bands. **73 de ON3VZ, Kristof**

Find the whole story online

Everything in this article is described in much more detail on my website, with the choices, the settings and above all the reasons why.

Website	on3vz.github.io
Blog	on3vz.github.io/blog
Logbook and map	on3vz.github.io/logbook
Shack	on3vz.github.io/shack
Portable	on3vz.github.io/portable
Mobile	on3vz.github.io/mobile
QSL wall	on3vz.github.io/qsl-wall
Awards	on3vz.github.io/certificates
QRZ.com	qrz.com/db/ON3VZ · qrz.com/db/ON3VZ/P
My club	WLD, ON6WL
RF.Guru case	My first HF station on RF.Guru

BLOG POSTS MENTIONED IN THIS ARTICLE

- Starting from Zero: Building a Capable HF Station on 25 Watts
- From First QSO to First QSL Card
- Eight Weeks On The Air: What 247 QSOs Taught a Class C Beginner
- Building the N1MM Field Day Tracker
- Building ON3VZ/P: a QRP Portable Station for Flora & Fauna
- 25 Watts to Borneo: My 15m SSB Contact with Indonesia
- Measuring is Knowing: the NanoVNA-H4 and tinySA
- When Theory Meets Practice: How a Layered Setup Keeps My 25 W Clean
- Digital Voice Modes at WLD with ON4PN
- My First Split QSO: Working an Iceland Pile-Up on 40 m
- IronWave vs HyEndFed: what each antenna does
- DM-1801 Pro, Part 1: Setup

73 de ON3VZ